

Euler–Euler simulation of multi-regime two-phase flows with thin liquid films

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Abstract

In Euler–Euler simulations, the information on the interfacial structure of two-phase flow is lost due to the underlying phase averaging. Suitable interaction force models, which rely on assumptions about the local interfacial structure, are required to close the two-fluid model equations. Recent advances have produced multi-regime closure models, which allow simulations of continuous-dispersed and separated flows within the same domain. Here, a model is presented that extends the multi-regime two-fluid model by a hybrid model for wall-bounded liquid films. Thus, wall films of varying film thickness and length scales can be considered in the Euler–Euler framework. The two-fluid film model (TFFM) is implemented in Star-CCM+ v15.04 and compared against experimental data for horizontal annular flow through an orifice and an expansion. It is found that the presented model can qualitatively reproduce the measured mean liquid volume fraction and film thickness profiles, including film separation and transition to stratified flow. Spurious entrainment of liquid in the gas core, however, deteriorates the quantitative agreement with the experimental data, which can be attributed to limitations in the current model implementation. Finally, the presented model is applied to the simulation of a water separator and compared to a simulation without a hybrid film model. A significant improvement in the predicted phase field and separation efficiency was found with the TFFM, which justifies further research to alleviate the current model limitations.

Keywords

thin film flow
multi-regime model
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1 Introduction

For engineering-type computational fluid dynamics (CFD) simulations of two-phase flows at length scales typically found in industrial applications, the Euler–Euler two-fluid model (TFM) is often adopted due to its comparably smaller computational cost. The flow is described statistically by averaging the underlying conservation equations for mass, momentum and energy before the discretisation. As a result of the averaging procedure, the microscopic phase interactions regarding mass and momentum require closure models which have to be selected appropriately for each flow morphology.

The TFM is often applied to the simulation of continuous-dispersed flows, where fluid particles are typically much smaller than the grid size, which is advantageous in terms of computational efficiency. Many authors have contributed

to the numerical method development for this type of flow (Frank et al., 2005; Ishii and Mishima, 1984) and a vast amount of closure models have been published in the last decades. The efforts in consolidating the various closure models (Hänsch et al., 2021; Li et al., 2023; Lucas et al., 2016) document the development for continuous-dispersed flows.

In practical applications, regime transitions and multi-regime flows are often observed, which has led to the development of multi-regime closure models for the TFM (Egorov et al., 2004; Höhne and Vallée, 2009). Recently, hybrid methods for the TFM have been developed (Coste, 2013; Gada et al., 2017; Mathur et al., 2019; Shonibare and Wardle, 2015; Wardle and Weller, 2013), which enable local application of sharp interface numerical schemes (without interface reconstruction) within the TFM framework. With these techniques, larger interfacial structures can be resolved, whereas subgrid-size structures are modelled. So

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far, these methods focus on transitions between separated and dispersed flows. Another type of small-scale interfacial structure is thin liquid films, which are relevant in various technical applications. For example, in evaporators, separators and bearings, the presence of wall-bounded liquid films affects the heat transfer, separation efficiency and wall shear stress, respectively. In a hybrid multi-regime TFM simulation, thin films can in principle be resolved by locally applying a sharp interface method similar to the volume of fluid (VOF) method (Wang et al., 2023). However, this requires a large number of mesh cells in the film region. It seems more appropriate in the averaged TFM approach to utilize a model for the subgrid-size liquid film.

Thin-film models have been applied previously in the simulation of spray-wall interactions in internal combustion engines (Foucart et al., 1998; Stanton and Rutland, 1998) and have also been used in hybrid modelling approaches. Adechy and Issa (2004) combined a thin film model with an Euler–Lagrangian description of droplet flow for the simulation of horizontal annular flow. The authors were able to reproduce experimental data for the fully developed state. Also for annular flow, Li and Anglart (2016) compared hybrid Euler–Euler and Euler–Lagrangian models for the core flow in combination with a thin film model and obtained comparable results with both modelling approaches. As thin films transition to separated flow, the film thickness may span several orders of magnitude. Consequently, Kakimpa et al. (2016) proposed a hybrid VOF and thin film model approach for the simulation of oil flow in a bearing casing. Depending on the local film volume fraction in wall-bounded computational cells, the film is either considered in the thin film model or in the VOF method. A similar approach was later implemented in the proprietary solver STAR-CCM+ by Ade (2019) and first applied to the simulation of vehicle water management. Such a hybrid modelling approach is also desirable for the Euler–Euler framework to extend the range of applicability to thin film flows. Here, such a hybrid modelling approach for multi-regime two-phase flow is presented, which combines a thin film model with a TFM for dispersed and separated flow morphologies. The combined model is validated for the case of developing annular flow through different pipe obstructions using previously published experimental data (Porombka et al., 2021). Finally, the applicability of the presented model to component-scale problems is demonstrated by simulating the flow in a droplet separator.

2 Modelling

2.1 Multi-regime two-fluid model

The hybrid modelling approach is based, in one part, on

the TFM for the description of local droplet, bubbly or large-scale interface flow regimes. The adiabatic transport of the interpenetrating gas and liquid phases is described in terms of the TFM mass and momentum equations (Ishii and Mishima, 1984):

$$\frac{\partial \alpha_k \rho_k}{\partial t} + \frac{\partial}{\partial x_i} (\alpha_k \rho_k u_{k,i}) = 0 \quad (1)$$

$$\begin{aligned} \frac{\partial}{\partial t} (\alpha_k \rho_k u_{k,i}) + \frac{\partial}{\partial x_i} (\alpha_k \rho_k u_{k,i} u_{k,j}) = \\ - \frac{\partial \alpha_k p}{\partial x_i} + \frac{\partial}{\partial x_j} [\alpha_k (\tau_{k,ij} + \tau_{k,ij}^T)] + \alpha_k \rho_k g_i + M_{k,i} \end{aligned} \quad (2)$$

The density ρ_k , velocity component $u_{k,i}$ of phase k and the pressure p are to be understood as the phase average of the corresponding microscopic variables. The phase fraction α_k is considered as a volume fraction here, $\tau_{k,ij}$, $\tau_{k,ij}^T$ denote the viscous and turbulent Reynolds stresses, g_i is a gravity component and $M_{k,i}$ denotes a component of the phase interaction force per unit volume. The turbulent stresses $\tau_{k,ij}^T$ are modelled with an SST k – ω eddy viscosity model in each phase (Menter et al., 2003). Turbulence model modifications due to multiphase flow are introduced at large-scale interfaces where the damping of turbulent fluctuations is accounted for by an ω –source term according to Tekavčič et al. (2021).

The phase interaction $M_{k,i}$ contains correlations of the unknown instantaneous variables and therefore has to be modelled. In continuous-dispersed flows, the interaction force is usually modelled as a linear combination of particle forces such as drag, shear lift or virtual mass (Liao et al., 2020). Here, a multi-regime closure model according to Höhne et al. (2020) is adopted, where regime-specific closure laws are blended locally. The blending can, in principle, be applied to any interaction force. Here, only the drag force is considered, which is written as a linear combination of drag models for droplet, bubbly and interfacial flow regimes

$$\mathbf{M}_D = (1 - |\psi|) \mathbf{M}_{D,surf} + \max(\psi, 0) \mathbf{M}_{D,drop} + \min(\psi, 0) \mathbf{M}_{D,bubb} \quad (3)$$

The local flow regime indicator function ψ , with the range of values $-1 \leq \psi \leq 1$, is defined as

$$\psi(\alpha_G, \nabla \alpha_G) = \varphi_{surf} (f_{bubb}(\alpha_G) - f_{drop}(\alpha_L)) \quad (4)$$

where the factors of ψ are the blending functions for the dispersed gas (bubb), dispersed liquid (drop) and large interface (surf) regimes according to

$$\varphi_{surf} = \frac{1}{2} \left(1 + \cos \left(\frac{\pi (\overline{\nabla \alpha_G} - (\nabla \alpha|_{\max} (1 - \delta_2)))}{2 \delta_2 \nabla \alpha|_{\max}} \right) \right) \quad (5)$$

$$f_{bubb} = \frac{1}{2} \left(1 + \cos \left(\frac{\pi(\bar{\alpha}_G - (\alpha_{d,max} - \delta_1))}{2\delta_1} \right) \right) \quad (6)$$

$$f_{drop} = \frac{1}{2} \left(1 + \cos \left(\frac{\pi(\bar{\alpha}_L - (\alpha_{d,max} - \delta_1))}{2\delta_1} \right) \right) \quad (7)$$

The modelling constants $\delta_1 = 0.05$, $\delta_2 = 0.1$, $n = 2$, $\alpha_{d,max} = 0.3$ and the normalized volume fraction $\bar{\alpha}_G$ and volume fraction gradient $\nabla \bar{\alpha}_G$ are adopted from Höhne et al. (2020).

A heat map of the blending function is given in Fig. 1 for the parameters as used in this work.

Essentially, droplet or bubbly flow is assumed where the dispersed phase volume fraction is below $\alpha_{d,crit}$ or $\alpha_{b,crit}$, respectively, and where the volume fraction gradient is

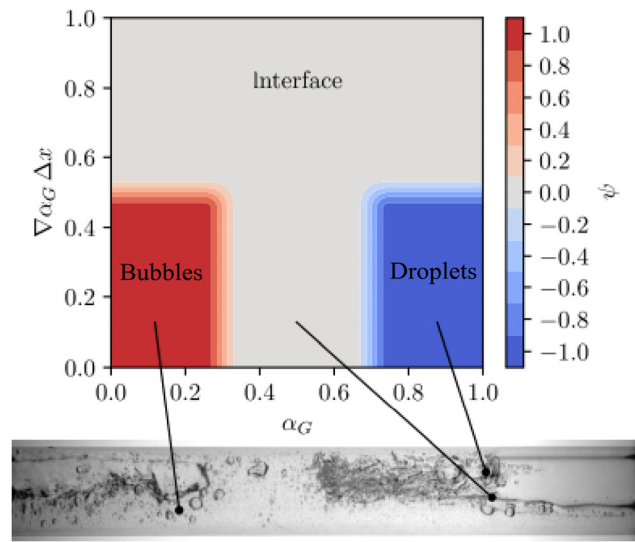


Fig. 1 Heat map of regime indicator function ψ with parameters $\alpha_{b,crit} = \alpha_{d,crit} = 0.3$, $\delta_1 = 0.05$, $\delta_2 = 0.1$, $n = 2$ (top) and illustrative slug flow image.

smaller than $1/(n\Delta x)$ with Δx being the local grid spacing. Otherwise, large-scale interface flow is assumed. For the bubbly and droplet regimes, the Schiller and Naumann (1933) particle drag model is assumed whereas the Strubelj and Tiselj (2011) linear drag is imposed at large interfaces. The drag model formulations and parameters are summarized in Table 1.

Additionally, the turbulent dispersion force (TDF) acting on the droplet phase is accounted for in a subset of the simulations. The TDF is modelled according to Burns et al. (2004) based on a spherical particle assumption. The source term acting on the dispersed phase (subscript d) due to the velocity fluctuations in the continuous phase (subscript c) is given in its modelled form as

$$\mathbf{M}_{TD,d} = -\mathbf{M}_{TD,c} = \frac{3}{4} C_D \frac{\alpha_d}{d_d} |\mathbf{u}_d - \mathbf{u}_c| \frac{\mu_c^t}{\sigma_{a,c}} \left(\frac{1}{\alpha_c} - \frac{1}{\alpha_d} \right) \nabla \alpha_c \quad (8)$$

and is added to Eq. (2). Here, C_D denotes the drag coefficient and d_d the dispersed phase characteristic diameter, μ_c^t is the dynamic eddy viscosity of the continuous phase and the turbulent Prandtl number for diffusion of volume fraction $\sigma_{a,c} = 1$ is adopted from Burns et al. (2004).

2.2 Thin liquid film model

The laminar thin film model available in STAR-CCM+ v15.04 is adopted here. It assumes that wall-bounded films are thin enough for the incompressible boundary layer approximations (Schlichting and Gersten, 2000) to be applied. The film flow is modelled by the height-averaged film mass and momentum equations, which are obtained from wall-normal integration of the boundary layer equations (Foucart et al., 1998). The mass conservation without mass transfer and the momentum conservation

Table 1 Drag force $M_{D,d}$ and interfacial area density a_i closure laws used in the multi-regime model, d – dispersed phase, c – continuous phase

Regime	a_i	$M_{D,d}$
		$\alpha_d C_D \frac{a_i}{4} \frac{\rho_c}{2} \mathbf{u}_c - \mathbf{u}_d (\mathbf{u}_c - \mathbf{u}_d)$, with
Bubble	$\frac{6\alpha_d}{d}$	$C_D = \begin{cases} \frac{24}{Re_d} (1 + 0.15 Re_d^{0.687}), & Re_d \leq 10^3 \\ 0.44, & Re_d > 10^3 \end{cases}$
Interface	1	$\frac{\alpha_c \alpha_d}{\tau} \rho_m (\mathbf{u}_c - \mathbf{u}_d)$, with $\tau = 10\Delta t$
Droplet	$\frac{6\alpha_d}{d}$	As for bubble regime



are given as

$$\frac{\partial h}{\partial t} + \nabla_s \cdot \mathbf{q} = \Gamma_{F,hf} + \Gamma_{F,sep} \quad (9)$$

$$\frac{\partial \mathbf{q}}{\partial t} + \nabla_s \cdot (\mathbf{q}\mathbf{q}) = -\frac{h}{\rho_F} \nabla_s p + \frac{\mathbf{t}_i - \mathbf{t}_w}{\rho_F} + \mathbf{g}h + \mathbf{M}_{F,hf} + \mathbf{M}_{F,sep} \quad (10)$$

with the surface gradient $\nabla_s = (\partial / \partial x, \partial / \partial y)^T$ and integral film velocity

$$\mathbf{q} = \int_0^z \mathbf{u} dz \quad (11)$$

Here, the film height is denoted as h , the wall-parallel film velocity as $\mathbf{u} = (u_x, u_y)^T$, the interfacial and wall shear stress vectors $\mathbf{t}_i, \mathbf{t}_w$, the gravity vector $\mathbf{g}$, the film density ρ_F and film pressure p which is equal to the imposed pressure from the external flow. The source terms $\Gamma_{F,hf}$, $\mathbf{M}_{F,hf}$, $\Gamma_{F,sep}$, $\mathbf{M}_{F,sep}$ are introduced to model hybrid film transfer and liquid film separation at sharp edges, respectively, and are discussed in the following sections.

The height-averaged equations are solved in a two-dimensional shell layer which replaces the solid-wall boundary faces of the volume mesh as shown in Fig. 2(a). Kinematic coupling between the film phase and the Eulerian phases is realized via volume partitioning $V_{cell} = V_F + V_G + V_L$ of cells in the first cell layer as shown in Fig. 2(b).

The volume fraction available to the TFM gas and liquid phases is defined in analogy to the porosity in porous media

$$\eta = \frac{V_G + V_L}{V_{cell}} = 1 - \alpha_F \quad (12)$$

where α_F denotes the film volume fraction. Further, the scaled volume fraction of the TFM gas and liquid phase is defined as

$$\alpha_k = \frac{V_k}{V_G + V_L} = \frac{\alpha_k}{\eta} \quad (13)$$

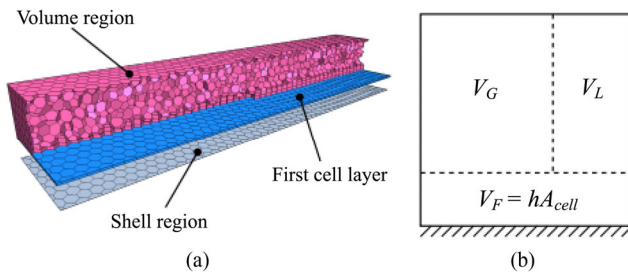


Fig. 2 Schematic of the shell and volume mesh region with the first cell layer (a) and volume partitioning of a finite-volume cell with a film interface (b).

so that

$$\sum_k \alpha_k = 1 \quad (14)$$

also in cells with a film interface. The TFM basic Eqs. (1) and (2) are then reformulated with respect to the available volume ηdV , i.e., the mass balance equation is stated as

$$\frac{\partial \alpha_k \eta \rho_k}{\partial t} + \frac{\partial}{\partial x_i} (\alpha_k \eta \rho_k u_{k,i}) = 0 \quad (15)$$

2.3 Film separation model

At sharp wall edges, the liquid film flow may partially detach from the edge depending on the local flow conditions. Here, an empirical correlation by Friedrich et al. (2008) is adopted for the separation criterion, which is based on a force ratio R between inertia, gravity and surface tension forces. In the film continuity equation, the source term

$$\Gamma_{F,sep} = \dot{x}_{sep} \frac{\rho_F}{h} \int_0^z \mathbf{u} \cdot \mathbf{n} dz$$

represents the volume per unit area of liquid film separated from the wall, where $\mathbf{n}$ denotes the edge normal vector. In STAR-CCM+, the separated film mass fraction $\dot{x}_{sep}$ is interpolated from the experimental data by the function in Eq. (16):

$$\dot{x}_{sep} = \begin{cases} 0, & R \leq R_c \\ 0.44(R - R_c), & R_c < R \leq (R_c + 1.6) \\ 0.057(R - R_c - 1.6) + 0.704, & (R_c + 1.6) < R \leq (R_c + 6.792) \\ 1, & (R_c + 6.792) < R \end{cases} \quad (16)$$

where R is a function of the local film Reynolds number, Weber number and the edge angle. The critical force ratio $R_c = 1$ is used in this work. A corresponding secondary momentum source term due to film separation $\mathbf{M}_{F,sep} = \Gamma_{F,sep} \mathbf{q}_F / h$ is added to the film momentum equation.

2.4 Hybrid film representation

In the liquid film model as implemented in STAR-CCM+ v15.04, the kinematic coupling of the film with the Euler-Euler phases is limited to the first cell layer. In multi-regime flow, the film thickness can however span several orders of magnitude, e.g., during annular to separated flow transition. Therefore, a hybrid model is proposed here, where the film flow is modelled either with the thin film model or with the two-fluid model.

The central idea is to switch the modelling framework based on the local ratio of film thickness and cell size. In the hybrid TFFM, the mass and momentum transfer between the thin film model and the two-fluid model is facilitated by adding explicit source terms $\Gamma_{F,hf}$, $\Gamma_{L,hf}$, $\mathbf{M}_{L,hf}$, $\mathbf{M}_{F,hf}$ to the respective mass and momentum equations. The gas and liquid phase mass and momentum equations of the TFFM are formulated in terms of the available cell volume ηdV and given as

$$\frac{\partial \alpha_k \eta \rho_k}{\partial t} + \frac{\partial}{\partial x_i} (\alpha_k \eta \rho_k u_{k,i}) = \Gamma_{L,hf} + \Gamma_{L,sep} \quad (17)$$

$$\begin{aligned} \frac{\partial}{\partial t} (\alpha_k \eta \rho_k u_{k,i}) + \frac{\partial}{\partial x_i} (\alpha_k \eta \rho_k u_{k,i} u_{k,j}) = & -\frac{\partial \alpha_k \eta p}{\partial x_i} + \frac{\partial}{\partial x_j} (\alpha_k \eta (\tau_{k,ij} + \tau_{k,ij}^T)) \\ & + \alpha_k \eta \rho_k g_i + M_{k,i} + M_{L,hf,i} + M_{L,sep,i} \end{aligned} \quad (18)$$

The film transport equations in the TFFM are given by Eqs. (9) and (10). $\Gamma_{L,hf}$ denotes the mass source per unit volume transferred from the thin film to the two-fluid model and $\mathbf{M}_{L,hf}$ is the corresponding secondary momentum transfer. As a criterion for the model transition, a simple thresholding based on a critical liquid volume fraction value in the first cell has been proposed by Kakimpa et al. (2016) and Ade (2019). This approach is also adopted in this work, so that the threshold value $\alpha_{F,t}$ is based on the total liquid volume fraction

$$\alpha_{L,tot} = \alpha_L \eta + \alpha_F \quad (19)$$

As is illustrated in Fig. 3, the total liquid mass is transfer from the two-fluid model to the thin film model when $\alpha_{L,tot} \leq \alpha_{F,t}$ and vice versa when $\alpha_{L,tot} > \alpha_{F,t}$.

In the two-fluid model, sudden jumps in field quantities are undesirable from a numerical point of view; thus, the mass transfer term in Eq. (17):

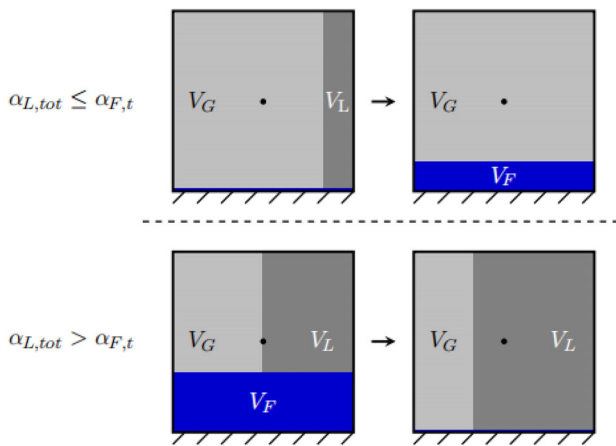


Fig. 3 Illustration of the model transition in a wall-bounded cell in the TFFM: transition from LFM to TFM when the threshold value is exceeded (bottom) or transition from TFM to LFM when the threshold value is not reached (top).

$$\Gamma_{L,hf} = \frac{\rho_L}{\tau_{hf}} (\alpha_{L,tot} - \alpha_{F,t}) \alpha_{avail} \quad (20)$$

is proportional to the offset from the threshold value. Here, the time scale for the model transition is related to the time step Δt via $\tau_{hf} = c_\tau \Delta t$. The model parameter $c_\tau \geq \max(\alpha_{L,tot}) - \alpha_{F,t}$ should be small and mainly depends on the numerical stability of the solver. The liquid volume fraction available for model transition

$$\alpha_{avail} = \begin{cases} \min(\alpha_F - \alpha_{F,min}, \alpha_{F,max}), & \alpha_{L,tot} > \alpha_{F,t} \\ \alpha'_L \eta, & \alpha_{L,tot} \leq \alpha_{F,t} \end{cases} \quad (21)$$

represents either the film phase or the liquid phase volume fraction. Two limiters, $\alpha_{F,min}$, $\alpha_{F,max}$ for the film volume fraction are introduced. The lower limit is introduced to ensure numerical stability of the film model with a residual film height $h_{min} = \alpha_{F,min} h_{cell}$. The upper limit corresponds to the maximum film volume considered in cell partitioning. Secondary momentum transfer is accounted for in Eq. (18) by

$$\mathbf{M}_{L,hf} = \begin{cases} \Gamma_{L,hf} \frac{1}{h} \dot{\mathbf{q}}_F, & \alpha_{L,tot} > \alpha_{F,t} \\ \Gamma_{L,hf} \dot{\mathbf{u}}_L, & \alpha_{L,tot} \leq \alpha_{F,t} \end{cases} \quad (22)$$

Corresponding source terms per unit area are added to the liquid film model equations:

$$\begin{aligned} \Gamma_{F,hf} &= -\Gamma_{L,hf} h_{cell} \\ \mathbf{M}_{F,hf} &= -\mathbf{M}_{L,hf} h_{cell} \end{aligned}$$

The parameters of the TFFM are summarized in Table 2.

Table 2 TFFM model parameters in validation cases

Parameter	Value
c_τ	2
$\alpha_{F,t}$	0.3
h_{min}	1 μm
$\alpha_{F,max}$	0.8

3 Experimental validation

3.1 Experimental setup

Verification and validation of the TFFM implementation are performed by comparing the simulation results with experimental data. To this end, the horizontal air–water pipe flow experiments of Porombka et al. (2021) are well suited for the model validation since spatially resolved volume fraction and film thickness data has been measured. An illustration of the experimental setup is given in Fig. 4.

The experimental setup consists of a spray-type air–water mixer followed by flanged acrylic pipe modules, which can

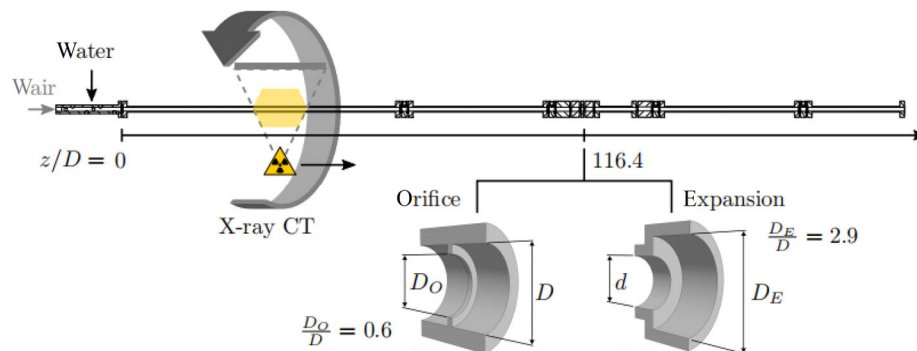


Fig. 4 Sketch of the horizontal test section with air–water mixer, orifice/expansion modules, and X-ray tomograph.

be combined with different internals and measurement modules. The pipe is suspended horizontally to pass through the aperture of the X-ray tomography system without introducing any additional attenuating material in the support volume. Alignment of the pipe with the horizontal is ensured within a maximum deviation of $\pm 0.1^\circ$. Here, two test section geometries are considered: A horizontal pipe with a thin orifice (case B) and a sudden expansion (case C). In both cases, the change in cross-section is located at $116.4D$ from the mixer outlet to ensure fully developed flow conditions at the obstacle. The main geometrical parameters are given in Table 3. The time-averaged 3D volume fraction and film thickness distributions have been measured at high spatial resolution from multiple X-ray micro-CT scans taken at consecutive axial positions. The axial range contained in the CT scans spans from $z/D = 104.3$ to $z/D = 129.2$ (case B) and $z/D = 134.2$ (case C), respectively. This includes the immediate upstream and downstream regions of the orifice or expansion. Stratified, wavy, slug, and annular flows have been investigated in the experiment. For the purpose of model validation, the operating point 4 is selected where annular flow is observed. The dimensionless numbers for these conditions are stated in Table 3, where the Reynolds numbers are based on superficial velocities, and the modified Froude number is given according to

$$Fr^* = \sqrt{\frac{\rho_G}{\rho_L - \rho_G}} \frac{U_G}{\sqrt{gd}}$$

Table 3 Geometrical parameters and dimensionless numbers for validation cases at operating point 4

Parameter	Case B (orifice)	Case C (expansion)
Axial distance from mixer	$\Delta z = 116.4D$	$\Delta z = 116.4D$
Area ratio	$(D_O / D)^2 = 0.6$	$(D_E / D)^2 = 2.9$
$Re_{s,G}$		25,000
$Re_{s,L}$		4090
Fr^*		2.53

In the fully developed region, annular flow with an asymmetric film thickness is observed in the experiment as would be expected for the given Reynolds and Froude numbers (Cioncolini and Thome, 2013). More details on the experimental setup and measurement techniques can be found in Porombka et al. (2021).

3.2 Simulation setup

The basic TFM and LFM equations are solved numerically in STAR-CCM+ v15.04. The multi-regime blending of the drag models according to Section 2.1 and the hybrid film representation according to Section 2.4 are implemented on the field function level using the Java API. For the incompressible simulations, the constant fluid properties are set to match the Reynolds and Froude numbers of the experiment given in Table 3. Fixed bubble and droplet diameters $d_{bubb} = 1$ mm, $d_{drop} = 160$ μ m, are used in the current simulation. The droplet diameter is set following the experimental results of Simmons and Hanratty (2001) and Hurlburt and Hanratty (2002), who measured the Sauter-mean droplet diameter in horizontal annular flow. At comparable superficial gas velocities, but at lower liquid velocities than in the current case, the measured mean diameters range from 80 to 130 μ m. Since the mean droplet diameter tends to increase with the liquid mass flux, a slightly larger droplet size was assumed for the current case. The bubble size was arbitrarily set as the local bubble flow regime is not relevant for the annular flow validation case.

An unstructured polyhedral mesh is used for spatial discretisation. The mesh is composed of a layer of hexahedral cells at the walls and prismatic cells in the core region. In the converging or diverging flow region next to the orifice or expansion, general polyhedral cells are used while a prismatic cell layer at the walls is retained. The mesh structure is shown in different sections in Fig. 5.

The computational domain covers a pipe length of $100D$ for case B and $71.5D$ for case C. At the inlet boundary of the computational domain, the measured mean liquid volume fraction profile according to Fig. 6 is imposed. It is obtained

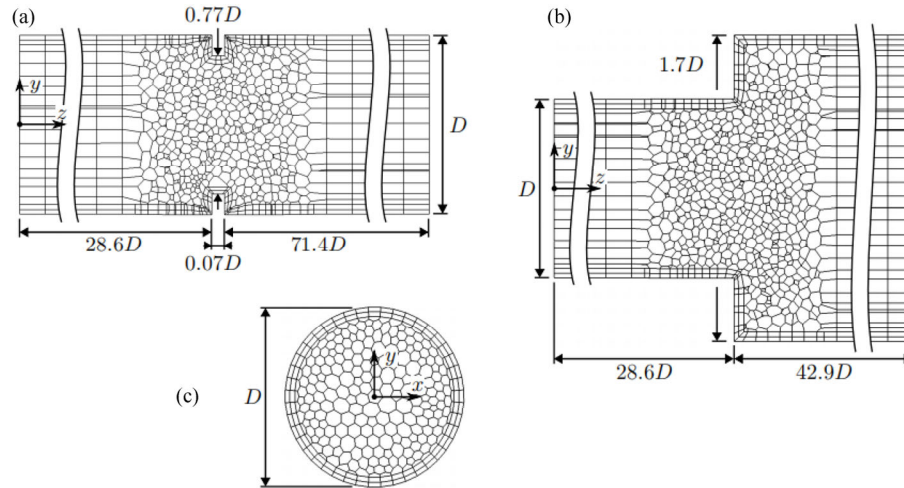


Fig. 5 Vertical sections through computational mesh for (a) case B, (b) case C and (c) cross-section for both cases.

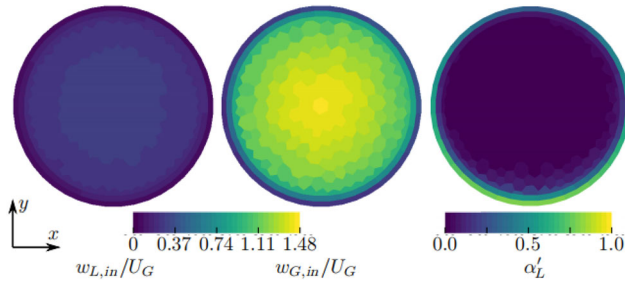


Fig. 6 Velocity and volume fraction profiles imposed on the inlet face of the computational domain for cases B and C.

from an axial average of the measured, time-averaged volume fraction profile in the fully developed region.

Since velocity data is not available from the experiment, the inlet velocity profiles are based on a precursor simulation of single-phase gas flow with equal $Re_{s,G}$ as in the experiment. With the resulting gas velocity profile in the fully developed region $w_{G,pre}(x, y)$ and the bulk liquid velocity W_L the mixture velocity profile at the inlet is defined as

$$w_{in}(x, y) = \alpha_G(x, y)w_{G,pre}(x, y) + \alpha_L(x, y)W_L \quad (23)$$

In a second step, the mixture velocity is scaled by a constant factor to obtain the gas and liquid inlet velocity profiles $w_{G,in}$, $w_{L,in}$ which reproduce the mass flow rates of the experiment. At the inlet edge of the shell region, the minimum film thickness h_{min} and zero film velocity are imposed. A constant pressure boundary condition is specified at the shell and volume region outlets, since the hydrostatic contribution in the case of annular flow is regarded as negligible. The solid walls are replaced by a shell region, which constitutes the 2D domain of the liquid film model. Whereas a no-slip condition at the walls is implicitly included in the film model, equal shear stress and pressure conditions are ensured at the interface by the solver (Siemens PLM, 2020).

3.3 Results for case B

In this validation case, horizontal annular flow is locally disturbed by a circular orifice, and the development of the liquid wall film is considered. The TFFM simulation results are compared to the experimental data in terms of the time-averaged liquid volume fraction and film thickness. The mean film thickness is obtained from the TFFM simulations analogously to the experiment by integrating the total liquid volume fraction, i.e., film model and two-fluid model contributions, in the wall normal direction (Porombka et al., 2021). The unsteady simulations are run for one second of physical time and the averaging of the gas, liquid and film phase variables starts at $t = 0.4$ s where the liquid hold-up reaches a steady state.

Figure 7 shows the mean volume fraction profiles in consecutive cross-sections along the pipe axis. The downstream edge of the orifice is located at the normalized axial coordinate $z^*/D = (z - z_0)/D = 0$. Based on the experiment (left column), the fully developed annular flow is dammed immediately upstream of the orifice, where the wall film accumulates. The wall film is stripped from the orifice edge and entrained in the gas core. Due to the film breakup, an annular droplet jet forms which is then re-deposited on the pipe wall and form a new wall film. At the most downstream position, the upstream annular flow profile is recovered. The TFFM simulation (center column) qualitatively reproduces the measured volume fraction profile and correctly captures the flow pattern, including the film separation at the orifice and re-deposition of entrained liquid. Quantitatively, a generally reduced liquid volume fraction is found in the simulation where the liquid holdup in pipe section contained in the CT scans is 0.037 compared to 0.096 in the experiment.

The reduced liquid holdup is related to limitations in

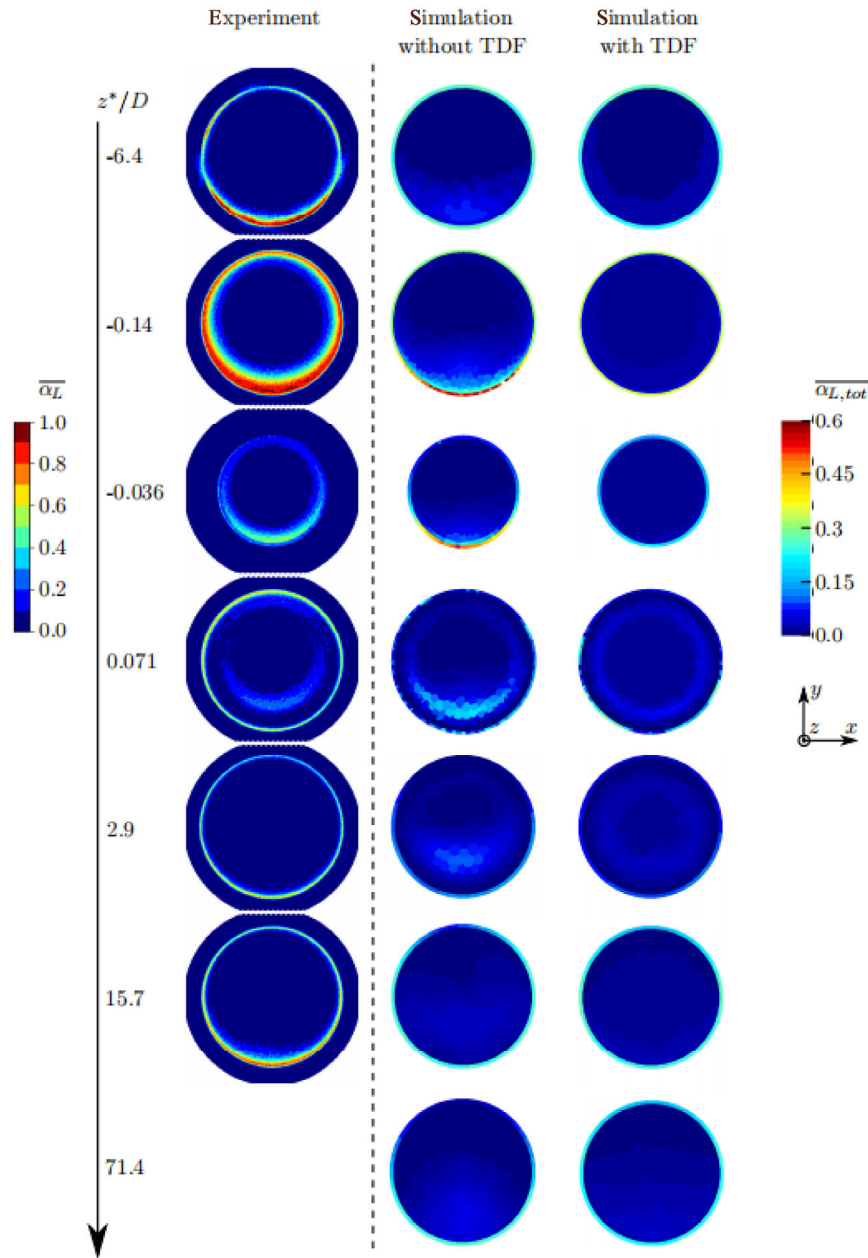


Fig. 7 Cross-sectional images of the time-averaged liquid volume fraction for case B at different axial positions (rows). Experimental data and TFFM simulation results with drag-only and drag + turbulent dispersion force (TDF) models (columns).

the hybrid film model implementation which are best understood by considering the hybrid film representation for case B in Fig. 8.

With the TFFM, the wall film is modelled either in the LFM or in the TFM, depending on the local total liquid volume fraction in the wall cells. The diagrams in Fig. 8 show the mean liquid film thickness in the LFM and the mean liquid volume fraction in the TFM on the pipe wall in a cylindrical coordinate system (r, φ, z^*) . The dashed line at $\varphi = \pi$ indicates the bottom of the pipe. On the inlet face, the wall film is initialised in the two-fluid model only,

and the modelling adapts further downstream according to the liquid volume fraction value at the wall. The film at the top of the pipe is mainly represented by the thin film model, i.e., $\bar{h} > 0$ mm and $\overline{\alpha_L} \approx 0$. Due to the action of gravity, the film thickness at the bottom of the pipe exceeds the threshold value, so that the liquid film mass and momentum are transferred to the two-fluid model. However, for any threshold value $\alpha_{F,t} \leq 1$ the corresponding film thickness is smaller than the first cell height that $\overline{\alpha_L} < 1$. Due to the finite transfer rates between the LFM and TFM both $\overline{\alpha_L}$ and $\bar{h}$ can adopt non-zero values.

Now consider the instantaneous representation of a film wave in the TFFM in Fig. 9, where the bottom diagram shows the film thickness in the LFM, the center image the liquid volume fraction in the TFM, and the top image the total liquid volume fraction.

The transition from the LFM to the TFM starts when the threshold volume fraction $\alpha_{F,t}$ is exceeded, which is visible at the right edge of the wave. At the onset of transition, the film thickness is smaller than one cell height, i.e., $\alpha_{F,t} < 1$ in the current implementation, since the coupling is limited to the first cell layer. The numerical schemes for volume fraction convection in the TFM cannot represent a sharp film at the initial subgrid-size thickness. Numerical diffusion at the highly sheared film interface leads to spurious liquid entrainment in the gas core, which consequently reduces the liquid hold-up and film thickness.

The diagrams in Fig. 10 show the mean liquid film thickness on the pipe wall in cylindrical coordinates for case B, from the experiment and the simulations. As was

discussed above, the mean film thickness in Fig. 10 is generally reduced compared to the experiment.

Qualitatively, the TFFM simulation reproduces the evolution of the wall film, including the complete separation at the orifice and the downstream reformation of the liquid film. This highlights the advantage of the model transition in the hybrid modelling approach. The film separation model (cf. Section 2.3) enables film stripping from the LFM to the TFM at the orifice.

The spreading of the entrained liquid downstream of the orifice is expected to be affected by turbulent dispersion. Accounting for this effect with a TDF model (cf. Section 2.1) influences the results for case B. Firstly, the film drainage due to gravity is counteracted by increasing the deposition of entrained liquid at the top of the pipe. This is visible by comparing the liquid volume fraction profiles at most downstream positions in Fig. 7. However, the additional diffusion of entrained liquid introduced by the TDF model also causes an additional decrease in liquid hold-up, which drops to 0.033. Further, the film thickness profiles upstream

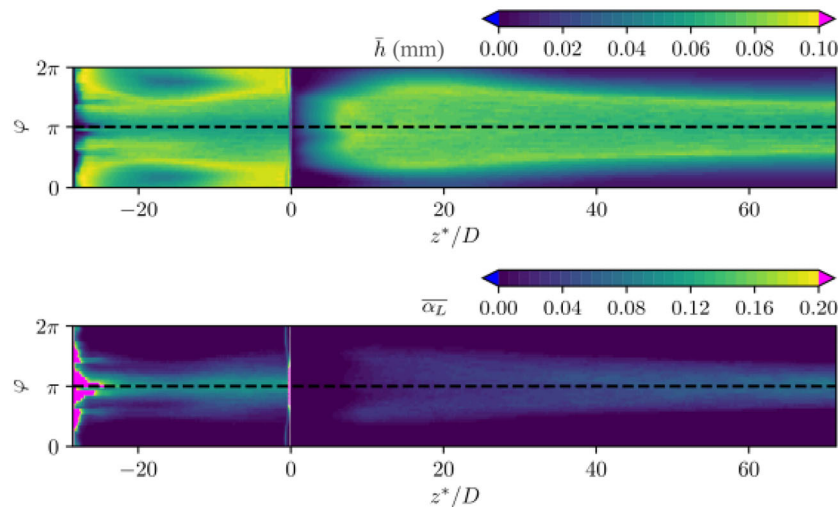


Fig. 8 Hybrid representation of the wall film in the TFFM simulation for case B. Mean film thickness $\bar{h}$ in the LFM (top) and mean liquid volume fraction $\bar{\alpha}_L$ in the TFM (bottom) on the pipe wall. The orifice is located at $z^* = 0$ and the dashed line indicates the bottom of the pipe.

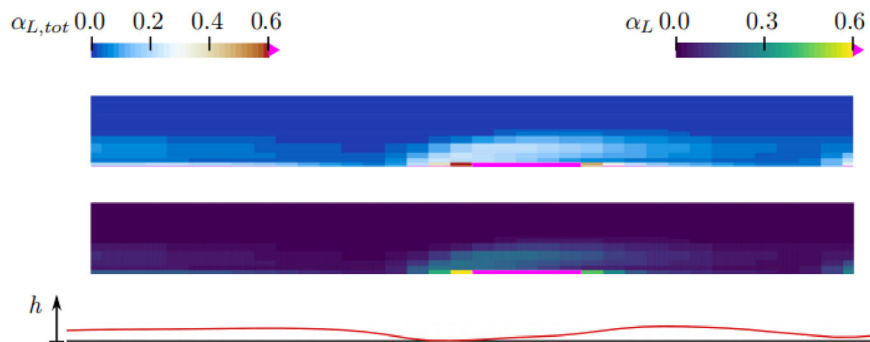


Fig. 9 Instantaneous film wave travelling from left to right in the central vertical section: total liquid volume fraction (top), liquid volume fraction (middle) and magnified film thickness (bottom).

of the orifice in Fig. 10 are significantly flattened when using the TDF model, so that the overall agreement with the experiment deteriorates.

3.4 Results for case C

In this validation case C, the developing horizontal annular flow through a pipe expansion is considered. The TFFM simulation of the flow regime transition from annular flow to stratified flow is compared to the experiment in terms of the mean liquid volume fraction and the film thickness. The flow pattern for case C is visualized in Fig. 12, which shows profiles of the measured liquid volume fraction at consecutive axial positions.

The annular flow on the upstream side separates from the edge of the smaller diameter pipe. The separated liquid film then breaks up into droplets so that a dispersed gas-liquid-jet is formed. Immediately downstream at $z^*/D = 0.071$ a liquid pool is visible at the pipe bottom,

and a liquid annulus forms at the larger diameter edge. The liquid phase accumulates here due to the flow reversal in the detached flow region. Due to the spreading of the jet, the liquid is deposited on the pipe wall, thus wetting the complete pipe circumference. Further downstream, the newly formed liquid film is drained, and a transition to stratified flow is observed.

As is highlighted in Fig. 12, the TFFM simulation can reproduce the film stripping to the continuous-dispersed morphology of the TFM at the edge of the expansion. Qualitatively, the flow regime transition from film to dispersed flow and to stratified flow in the larger diameter pipe is reproduced. However, the liquid volume fraction is generally reduced compared to the experiment, which is due to the spurious entrainment described in the previous section. Quantitatively, the liquid hold-up in the scanned pipe section is 0.015 in the simulation compared to 0.08 in the experiment. Consequently, the mean film thickness in Fig. 11 is reduced compared to the experiment. Opposed to

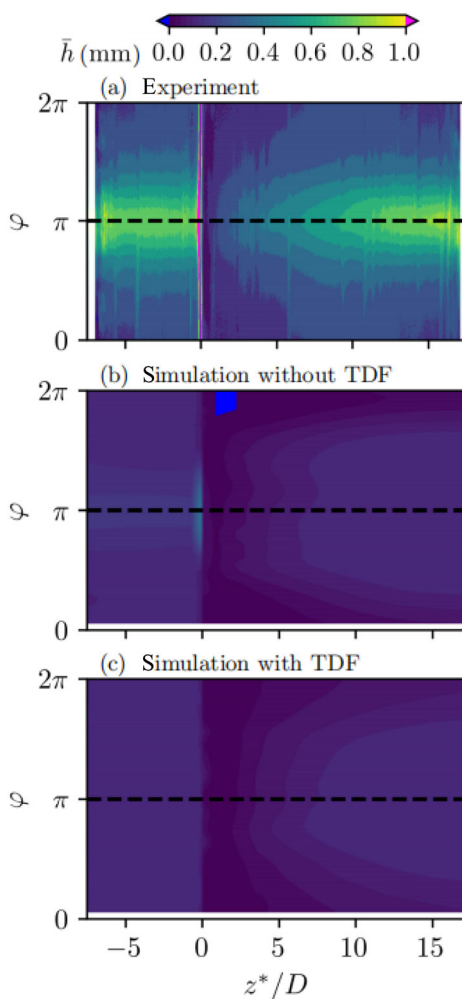


Fig. 10 Case B (orifice): Heat map of mean liquid film thickness on the pipe wall as measured in the experiment and from the TFFM simulation. Negative values are post-processing artefacts.

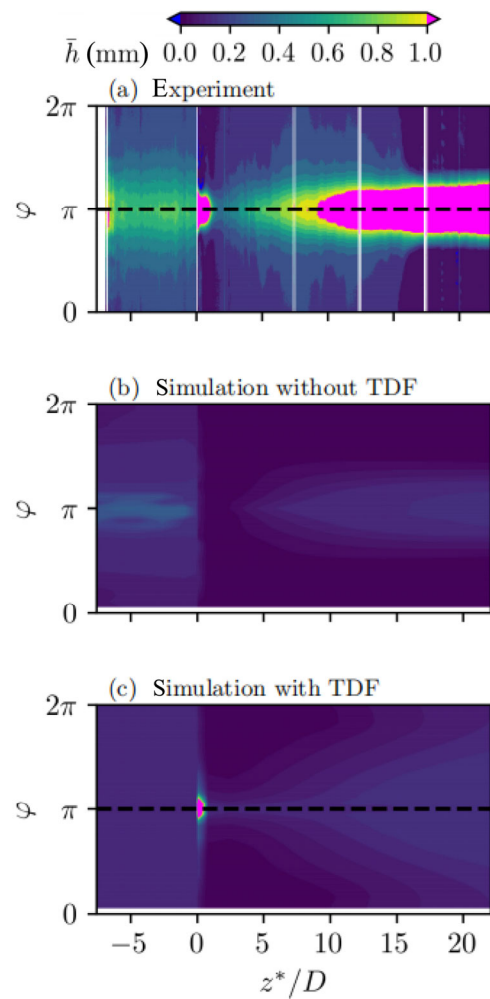


Fig. 11 Case C (expansion): Heat map of mean liquid film thickness on the pipe wall as measured in the experiment and from the TFFM simulation.

the experiment, the droplet deposition and film formation is localized in the bottom half of the pipe since the gas-droplet-jet resembles a ballistic curve. When the TDF model is included in the simulation, both the liquid pooling in Fig. 12 and the wetting of the top half of the pipe in Fig. 11 can be reproduced similarly to the experiment. Further, the liquid hold-up increases from 0.015 to 0.019 when the TDF model is applied and thus is shifted towards the measured value. These observations indicate that the TDF needs to be accounted for in the TFFM simulation to qualitatively reproduce the flow pattern downstream of the expansion. In spite of the observed reduced liquid hold-up for the case of annular flow, the TFFM can already

yield reliable predictions and improvement over the TFM in applications with less interfacial shear, as shown in Section 4.

4 Application and model comparison

In this section, we demonstrate the relevance of the presented model by applying it to a water separator with a generic geometry similar to water separators used in the industry. A model comparison shows the clear necessity of employing the hybrid film model to capture multi-scale liquid wall films, which cannot be realistically represented by a two-fluid model approach. It is noted here that the

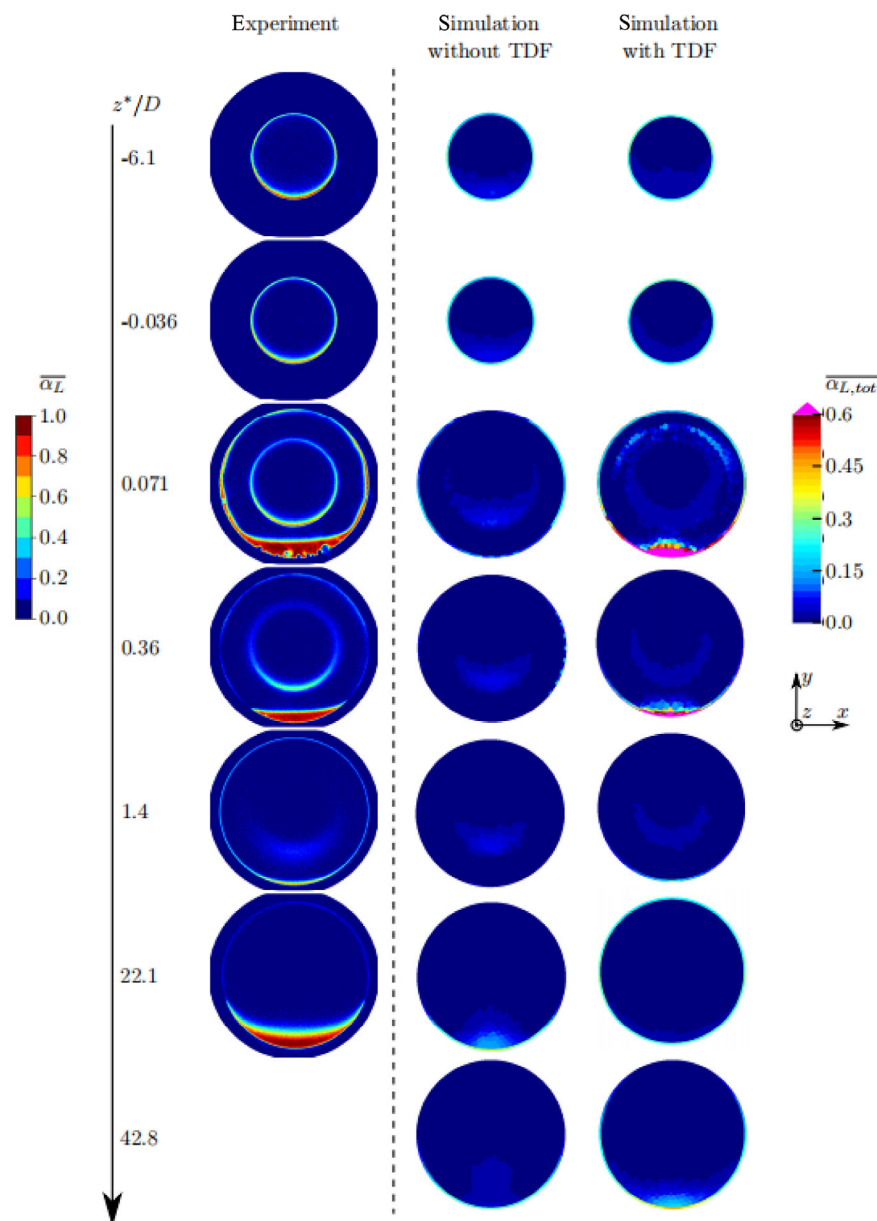


Fig. 12 Cross-sectional images of the time-averaged liquid volume fraction for case C at different axial positions (rows). Experimental data and TFFM simulation results with drag-only and drag + turbulent dispersion force (TDF) models (columns).

TFFM implementation for this application case deviates from the method which is presented in Section 2 in some details, which improve the numerical stability of the model. Specifically, the transition time scale factor c_τ and the film volume fraction limiter $\alpha_{F,max}$ are adjusted. Furthermore, polydispersity of the liquid phase is accounted for by a presumed log-normal distribution population balance approach. However, the fundamental models remain the same, and this case only has the purpose of illustrating the need for a hybrid film model.

4.1 Simulation setup

The geometry of the water droplet separator and the employed boundary conditions are shown in Fig. 13. Water droplets are separated from the air–water mixture by impinging on the surface of four rows of cylindrical obstacles.

At the inlet, constant air and water flow rates are set which are representative of typical fuel cell water separators found in the automotive industry. The development of the liquid wall film is a result of the TFFM simulation, and there is no film flow imposed at the inlet. Water is specified as a dispersed phase in the two-fluid model with a Sauter-mean droplet diameter of 250 μm . The simulation domain has two pressure outlets, a main air outlet and a second water outlet for drainage of the separated liquid. By using appropriate pressure drop conditions at the outlets, the air flow at the secondary outlet is adjusted to 10% of the total flow rate to facilitate the drainage of water. In a typical separator, this split ratio would be smaller. However, the specific choice is made here to illustrate the effect of the modelling on the separation efficiency rather than the effect of the split ratio. The computational mesh is built of polyhedra with a base size of 1.8 mm. Local refinement and two prism layers at the wall boundaries are used which results in an overall cell count of 2.3 million cells. Transient simulations are run until a quasi-steady state is reached with regard to the rate of change in the liquid hold-up and the outlet flow rates. The simulations are performed with the presented TFFM and with the MTFM according to Section 2.1. In the latter case a liquid film model is not included.

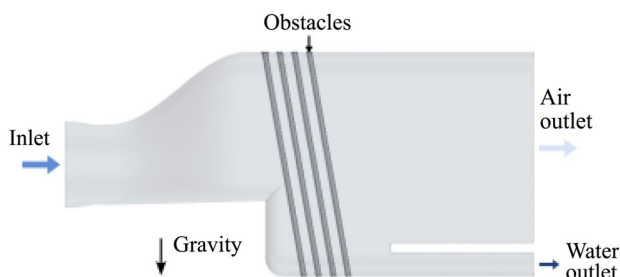


Fig. 13 Geometry of the generic separator with obstacles, inlet and outlet faces.

4.2 Results

Figure 14 shows the liquid mass fraction in the MTFM and TFFM simulations in the steady state. Visual inspection of the phase fields shows a large difference between both modelling approaches. With the MTFM in Fig. 14(a) only weak interaction of the droplet phase with the obstacles is observed as liquid accumulates in the wake regions. The large interface regime of the MTFM is not applied as the spatial resolution is not sufficient to resolve the thin liquid films on walls with the Eulerian–Eulerian approach. Consequently, the predicted separation efficiency is approximately 0%. With the TFFM in Fig. 14(b) the dispersed liquid phase impinging onto the obstacles is collected in the fluid film. Part of the film flows downward along the obstacles, until the liquid reaches the bottom. Part of liquid films is also stripped from the obstacles but the predicted droplet sizes of stripped liquid is larger which leads to a downward motion of those drops. The TFFM is required to capture the expected liquid pooling at the bottom of the separator. Quantitatively, the predicted separation efficiency increases to the expected percentage of approximately 90%. With a thin film model only, the implied boundary layer assumptions for the film would be violated where the local film thickness significantly exceeds thickness of wall bounded cell. Only with a hybrid film modelling approach, such multi-scale films are predictable at reasonable mesh sizes. To perform simulation of this type of flow with a VOF-like interface tracking technique including a regime detection is theoretically possible albeit at much finer mesh resolutions.

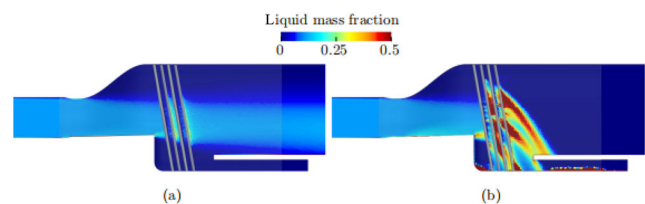


Fig. 14 Liquid mass fraction in vertical section through generic separator: (a) multi-regime two-fluid model and (b) TFFM.

5 Summary and conclusions

An Euler–Euler model for the CFD simulation of multi-regime gas–liquid flows including multi-scale wall bounded liquid films has been presented in this work. Various relevant technical applications motivate this development, e.g., the simulation of two-phase flows in heat exchangers, liquid separators or pipe lines. The so called TFFM is based in one part on the Euler–Euler two-fluid model with a morphology-adaptive closure model for droplet flow, bubbly flow and large scale interfaces. In the other part the model is based on a thin film model for wall bounded thin

liquid films. Both models are coupled by explicit volumetric source terms added to mass and momentum equations. Based on a local volume fraction threshold, wall-bounded film flow is either represented in the thin film or the two-fluid model. The presented model was implemented in the proprietary CFD solver Star-CCM+ v15.04 and is validated by comparing against measured liquid volume fraction and film thickness data for horizontal annular flow through an orifice or expansion. In both cases, the presented model is able to qualitatively reproduce the experimental data. It was found that the turbulent dispersion force needs to be included in the model to sustain the wall wetting at the top of the pipe and to reproduce the flow pattern downstream of the pipe expansion. Quantitatively, the liquid volume fraction and film thickness is generally reduced compared to the experiment due to spurious entrainment of liquid in the gas core. This is caused by the limitation of the film model and two-fluid model coupling to a single cell layer in the current implementation in combination with numerical diffusion in the volume fraction convection. Compressive numerical schemes for the volume fraction convection can potentially alleviate the spurious entrainment. Alternatively, the model coupling could be extended to multiple cell layers. It is however difficult to implement this non-local operation in the framework of the proprietary solver used in this study. Finally, the advantage and relevance of the hybrid film modelling approach in the TFFM is demonstrated using the example of a generic water separator as used in the industry. The comparison to a multi-regime two-fluid model without a thin film model shows a large difference in the predicted separation efficiency and the qualitative phase distribution. The presented model is able to predict a plausible result whereas zero separation efficiency is observed with the standard approach. Based on the validation and application results, further development of the presented model appears worthwhile to overcome the identified limitations, e.g., for highly sheared film flows with a film thickness in the order of the first cell height. The presented model already constitutes a relevant extension of the multi-regime two-fluid model in situations where film flow may occur.

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Declaration of competing interest

The authors have no competing interests to declare that are relevant to the content of this article.

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